$IG_T = 1 - 0.342$

$IG_T = 0.658$

B) T

T	Total	Yes	No	Entropy
Hot	2	1	1	1
Mild	3	2	1	
Cool	3	1	2	

Hot Calculate Entropy

$= -\frac{1}{2}\log_2\frac{1}{2} - \frac{1}{2}\log_2\frac{1}{2}$

$= -0.5\log_2 0.5 - 0.5\log_2 0.5$

$= -0.5\frac{\log(0.5)}{\log(2)} - 0.5\frac{\log(0.5)}{\log(2)}$

$= -0.5(-1) = 0.5(-1)$

$= 1$

(2) Info gain Entropy Calculation

outlook

outlook	Total	Yes	No	Entropy
Sunny	3	0	3	0
Cloudy	2	2	0	0
Rainy	3	2	1	0.9141

Entropy Calculate

$$\text{Rainy} = -\frac{2}{3}\log_2\frac{2}{3} - \frac{1}{3}\log_2\frac{1}{3}$$

$$= -0.66\log_2 0.66 - 0.33\log_2 0.33$$

$$= -0.66\,\frac{\log(0.66)}{\log(2)} - 0.33\,\frac{\log(0.33)}{\log(2)}$$

$$= -0.66(-0.59) - 0.33(-1.59)$$

$$= 0.9141$$

Weighted Entropy :-

$$= \frac{3}{8}(0) + \frac{2}{8}(0) + \frac{3}{8}(0.9141)$$

$$= 0 + 0 + 0.375(0.9141)$$

$$= 0.342$$

CamScanner

23122232

outlook Temperature Hot (1)

OL	T	H	W	R
Sunny	Hot	High	weak	No
Sunny	Mild	High	Strong	No
Cloudy	Hot	High	weak	yes
Rainy	mild	high	weak	yes
Rainy	Cool	Normal	weak	Yes.
Rainy	Cool	Normal	Strong	No
Rainy	Cool	Normal	Strong	Yes.
Cloudy	mild	Normal	Strong	Yes.
Sunny	Cool	Normal	weak	No

Sol

Data Prime attribute R

Yes = 4

No = 4

Entropy :-

$$= -\frac{4}{8}\log_2\frac{4}{8} - \frac{4}{8}\log_2\frac{4}{8}$$

$$= -0.5(\log_2 0.5 - 0.5\log_2 0.5$$

$$= -0.5\,\frac{\log(0.5)}{\log(2)} - 0.5\,\frac{\log(0.5)}{\log(2)}$$

$$= -0.5(-1) - 0.5(-1)$$

$$= 1$$